

Article - Engineering, Technology and Techniques

Business and Data Intelligence to Support the Construction Management Process in Electrical Power Utilities

Vitor Augusto Fiates Silva¹
<https://orcid.org/0009-0006-0509-7931>

Rafael Martins¹
<https://orcid.org/0000-0002-0776-2978>

Allan Said Fadel¹
<https://orcid.org/0009-0008-9771-6596>

Clailton Leopoldo da Silva¹
<https://orcid.org/0000-0001-7840-0352>

Mauro José Bubniak¹
<https://orcid.org/0009-0003-4657-9767>

Isabel Canalli Travagin¹
<https://orcid.org/0009-0000-0407-9842>

Bruna Machado Mulinari²
<https://orcid.org/0000-0001-5838-6178>

Clayton Hilgemberg da Costa²
<https://orcid.org/0000-0002-7117-1785>

Dierli Maiara da Rosa Maschio²
<https://orcid.org/0000-0002-1268-0790>

Flavio Lori Grando²
<https://orcid.org/0000-0002-2563-4953>

Germano Lambert Torres²
<https://orcid.org/0000-0003-3789-4696>

Jocelleide Dalla Costa Mumbelli²
<https://orcid.org/0000-0003-0511-5811>

Fabio Alessandro Guerra²
<https://orcid.org/0000-0001-8429-2263>

Milton Pires Ramos^{2*}
<https://orcid.org/0000-0002-9244-2417>

¹Copel - Companhia Paranaense de Energia, Curitiba, Paraná, Brasil; ²Gnarus Institute, R&D Department, Itajubá, Minas Gerais, Brasil.

Editor-in-Chief: Alexandre Rasi Aoki
Associate Editor: Alexandre Rasi Aoki

Received: 03-Dec-2024; Accepted: 18-Oct-2025

*Correspondence: miltonpiresramos@gmail.com, Tel.: +55-41-98402 3423 (M.P.R.).

HIGHLIGHTS

- Challenges faced in project and construction management in the energy sector.
- Data Analytics and Business Intelligence to support the management process.
- Improvements in decision-making, efficiency and reliability in construction management.

Abstract: The main objective of this work is to present a methodology and software tools that supports the efficient management of transmission line projects through integrated reports from various data sources and systems. This enables the management team to monitor project performance and make informed decisions to maximize overall returns for the company. To achieve this, the research adopts a business intelligence and data analysis-based approach, which includes mapping and organizing information from internal systems, creating a standardized data structure, and implementing managerial reports on a BI platform to monitor resource allocation and efficiency. As outcomes, the study presents a set of dashboards and reports

offering an integrated and comprehensive view of activities, including physical and financial progress indicators, risk analysis, and optimization in human resource allocation, highlighting significant savings in project budgets. This approach (methodology and tools) facilitates more accurate and timely analysis, enhances decision-making, efficiency and reliability, and provides strategic advantages for concessionaires in terms of competitiveness, operational sustainability and greater control over the resources employed in transmission projects.

Keywords: Construction management; power companies; business intelligence; data analytics.

INTRODUCTION

Electric power companies in Brazil, particularly those operating in the transmission sector, face growing challenges in managing their projects. The economic, social and strategic importance of electric power today is undeniable, and the need to expand and modernize infrastructure requires the execution of complex and large-scale projects. Such projects, which involve expansions, reinforcements, or improvements, present specific challenges such as the integration of various stakeholders (suppliers, contractors, public agencies), the management of land and socio-environmental issues, management of multidisciplinary teams and ensuring the safety of the electrical system during project execution. In Brazil these projects must also be compliant with Module 3 of the Rules for Electric Power Transmission Services of ANEEL (National Electric Energy Agency) [1].

Effective strategic decision-making in this scenario relies on accurate data and robust analyses. However, the dispersion of information across different systems and spreadsheets, the lack of standardization, and the difficulty in integrating relevant data hinder the ability to obtain a clear view of project performance. Consequently, decisions are often based on perceptions and intuitions, which can jeopardize resource optimization, cost control, and adherence to deadlines.

In light of this issue, this work proposes methodologies and software tools that leverage data intelligence combined with business intelligence to support the efficient management of projects in the electric power sector. These methodologies aim to organize, process, and qualify data, utilizing business intelligence platforms to provide managers with relevant and reliable information for strategic decision-making, particularly in resource allocation. Two application cases are presented in order to demonstrate and validate the use of both methodology and its software tool: project teams' allocation and capital expenditure follow up (CAPEX).

The developments presented in this article are part of the R&D project ANEEL PD-0641-0565/2019, titled "Process Mining with Dynamic Flowshop Scheduling Optimization under Uncertainty for Project Management," carried out by Instituto Gnarus in partnership with Copel Geração e Transmissão S/A. The following sections sequentially present the literature review, methodology, results, and conclusions.

RELATED WORKS

This section presents a review of bibliographic references, identifying the main publications on the researched topic, thereby forming a theoretical foundation for this study.

Lopes and Boscarili [2] conduct an investigation into the application of Business Intelligence and Analytics (BIA) tools in the construction sector. They emphasize the importance of management in this industry to meet project requirements, including business goals and stakeholder expectations. The study also highlights the potential of BIA tools to provide crucial managerial information for businesses, with significant applications in areas such as cost control, budgeting, workplace safety, waste management, and risk management, among others. This research identified numerous BIA tools suitable for the construction sector, each with different applications and potential benefits for improving organizational management within the industry.

The authors also observed that most BIA solutions were specifically developed for each case study due to the unique characteristics and needs of the construction industry. They emphasize the need for collaboration among companies, professional associations, and academic institutions to promote the widespread adoption of BIA tools in construction. In summary, the article highlights the importance of Business Intelligence and Analytics tools as partners in optimizing management within the construction sector, offering valuable insights for professionals and researchers aiming to enhance management practices in the field.

In their research, Gupta and Cannon [3] emphasize the importance of a data strategy in guiding business strategy for the digital transformation of organizations. The effectiveness of this strategy involves the systematic development of methods to collect, store, manage, and analyze data relevant to the organization's

operations and strategic objectives. This approach not only enables a deeper and more comprehensive understanding of current operations but also empowers leaders to forecast future trends and identify market opportunities.

The implementation of a data-driven strategy requires investments in robust technology and personnel training to ensure that data is accurately collected, of high quality, and available for timely analysis. Additionally, the authors discuss the various dimensions of data quality and their fundamental role in determining the reliability of insights, minimizing errors and biases that could distort conclusions.

The literature review shows that the terms "business intelligence" and "business analytics" are often used in the fields of data analysis and data science. However, the relationships and distinctions between these terms are not widely understood, and definitions vary among authors. Chae and Olson [4] consider business analytics and business intelligence as similar terms, both reflecting the use of analytical capabilities for decision support. In contrast, Wixom, Yen, and Relich [5] define business analytics as a process and business intelligence as insights, with business intelligence being derived from business analytics.

Sharda, Asamoah, and Ponna [6] categorize business analytics into three levels: descriptive, predictive, and prescriptive. Descriptive analytics aims to understand current trends and causes in business, focusing on what happened in the past and what is happening in the present. Predictive analytics uses statistical methods, such as machine learning, to forecast future events based on available information. Prescriptive analytics, on the other hand, provides actionable guidance for business decisions, using data derived from both descriptive and predictive analyses.

Chater and coauthors [7] clarify that data-driven decision-making involves inferences drawn from the information contained in the dataset. These inferences can be made automatically, using heuristics to identify and correlate recognizable patterns, or through the rational choice model, where the decision-maker follows a structured process aligned with organizational goals and established rules.

Regarding data strategy and data-driven decision-making, Loshin [8] emphasizes the importance of data quality in these scenarios, while defining key aspects to ensure an adequate use of data in projects. Marshal [9] goes even further, assessing that low quality data can make the process of making decisions in a business intelligence scenario extremely difficult or, in some cases, even impossible, while Debbarma [10] describes attributes to measure data quality, such as completeness, consistency, validity, integrity, conformity and accuracy.

Therefore, the aforementioned studies reinforce the idea that the widespread use of data analysis tools and business intelligence techniques has significantly transformed internal management processes within organizations, providing a more detailed and accurate analysis of business data. This advancement enables more informed and agile decision-making, encouraging organizations to rethink and improve their data strategies to align them with their business strategies. By developing synergy between data analysis and corporate objectives, companies can identify market opportunities, optimize operations, and improve customer satisfaction, resulting in a significant increase in performance and market competitiveness.

Although the cited texts gave our research good directions to follow, in none of these cases was the final application focused on the construction management in electrical power utilities. The hints were valuable and well used, but in the case of this research it was necessary to address some specificities related to the field such as safety, reliability, uninterrupted service, multidisciplinary teams and environmental issues.

METHODOLOGY

The methodology implemented for the development of the project and construction management support solution was developed in five main stages: mapping, modeling, development, validation, and delivery. Figure 1 provides a summarized overview of the main activities carried out in each stage of the project, which are detailed in the topics that make up this chapter.

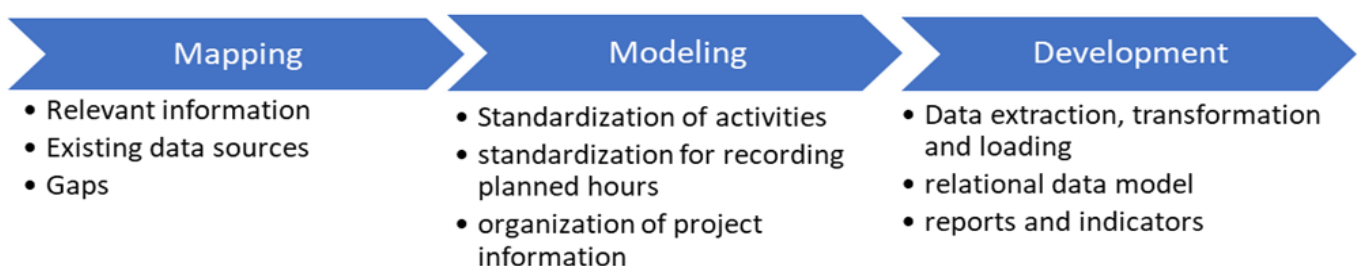


Figure 1. Solution development steps.

Mapping Essential Information and Existing Data Sources

During the mapping stage of relevant information to the project management process, interviews were conducted with the departments responsible for planning and implementing projects and construction at Copel Geração e Transmissão (Copel GeT). In the interviews and meetings with the teams, the essential information was identified and are listed in the first column of Table 1.

After mapping the essential information, meetings were held with the technical teams to identify the existing data sources within the company, such as software and control spreadsheets, that could contribute to the construction and consolidation of the desired information. During this stage, the data sources listed in Table 1 were mapped.

Table 1. Mapped essential information and respective data sources.

Essential Information	Data Sources
Project Data (Project Name, Description, Manager)	MS Project (Project Schedule) and Word (Project Opening Term)
Completion Date (Planned and Actual)	MS Project (Project Schedule) and MS Excel (Control Spreadsheet)
Physical Progress (Planned vs. Actual)	MS Project (Project Schedule) and MS Excel (Control Spreadsheet)
Financial Progress (Planned vs. Actual)	MS Excel and SAP ECC
Contracts (Total Value, Contract Balance, Addendums)	SAP ECC
Cost Collectors (budget planning)	SAP ECC
Project Risks	MS Excel (Risk Matrix Spreadsheet)
Man-Hour Allocation (Planned vs. Actual)	MDO (Internal labor settlement system) and SAP ECC

It is important to highlight that during this mapping stage, it was identified that the data was dispersed across various files and systems, creating complexity in the management of projects and the overall portfolio. This finding highlighted the need for implementing a solution with an organizational framework.

Gap Mapping

During the data source mapping phase, several gaps were identified that needed to be addressed to ensure that the solution for supporting project and construction management fully covered the scope of essential information for the management process. The identified gaps were:

- Lack of standardization in the identification of activities recorded in the worked hours log, meaning the same activity had multiple descriptions.
- Absence of a data source for recording planned activities for each employee, making it impossible to reconcile planned versus actual man-hours for each project.
- Need for the implementation of an organizational framework for project data.
- Need for the implementation of a preliminary data processing procedure.
- Need for the implementation of a validation and consistency checking process for the worked hours logs.
- Need for the implementation of standardized reports to monitor projects and the portfolio.

For each of the identified gaps, solutions were implemented, as described in the following sections.

Standardization of Activities

To address the issue of the lack of standardization in the identification of activities recorded in the hours control, meetings were held with the technical teams from planning and implementation at Copel GeT, and a new procedure for identifying activities was established. The identification of each activity was segmented into two levels, as outlined below.

Level 1 Activities

To consolidate the generic actions associated with each activity, the concept of Level 1 was established. The actions identified for this level of description include: analysis, preparation, inspection, management, coordination, supervision, and participation.

It is important to highlight that, with the generalization of this level, each department within the company involved gains a clear understanding of the meaning of each item listed. For example, in the case of the item "inspection," the department responsible for it may handle part or the entirety of the inspection activity, while another department may not have this item within its scope. In general, departments are not required to cover all the items listed at Level 1, nor are they fully responsible for them. For this reason, the need to create Level 2 for each activity was identified.

Level 2 Activities

To group the specific actions of the activities, Level 2 was created. The actions mapped for this level include: administrative (non-technical), administrative (technical), as-built, commissioning, contract, contract/terms of reference and related, technical documents (including projects), due diligence, equipment and materials, studies, inspection, materials, construction, R&D, civil or geological or land project, electrical project, electromechanical project, mechanical project, proposals, reports or schedules, responses to official letters and similar, meetings, company routine, and training.

Thus, it was defined that each activity would be composed by the combination of the terms from Level 1 and Level 2, thereby characterizing the activity itself. For example:

- Level 1: analysis;
- Level 2: contract;
- Resulting activity: Contract analysis.

Implementation of the structure for recording planned hours

To address the issue of missing data regarding the planning of hours to be worked by employees in the execution of project and construction activities, meetings were held with the technical teams, and a record-keeping structure was implemented through the development of a standardized Excel spreadsheet. This spreadsheet was replicated for each area and stored in a shared repository on SharePoint, as shown in Figure 2.

The screenshot shows a SharePoint interface for a private group named 'SIP/DGPI'. The left sidebar contains navigation options: 'Página Inicial', 'Conversas', 'Documentos', 'Compartilhado conosco', 'Bloco de Anotações', 'Páginas', 'Conteúdo do site', 'Lixeira', and 'Editar'. The main area displays a list of documents under the path 'Documentos > Dashboards > x - EXCLUSIVO SEP - não mexer'. The list includes columns for 'Nome', 'Modificado', and 'Modificado por'. The following table represents the data shown in the screenshot:

Nome	Modificado	Modificado por
DGET-Infos.xlsx	15 de agosto de 2022	Joachim Graf Neto
DGIP.xlsx	27 de outubro de 2023	Vitor Augusto Fiates Silva
DPMO.xlsx	27 de outubro de 2023	Vitor Augusto Fiates Silva
SEI-Base-Horas.pbix	21 de fevereiro	Allan Said Fadel
VEEM - Cópia.xlsx	17 de fevereiro de 2023	Nilson Marcelo Bonato
VEEM.xlsx	27 de outubro de 2023	Vitor Augusto Fiates Silva

Figure 2. Repository of planned timesheets.

The developed spreadsheet consists of fields that allow area managers and their supervisors to record manually the planned, scheduled activities for each employee and the estimated effort hours for their completion. The activities recorded in the planning spreadsheet were standardized to comply with the description and composition rules presented in the previous section. Figure 3 shows a sample of the planned hours tracking spreadsheet for the engineering team, which includes a list of employees, the activities to be performed by each one, as well as the anticipated effort for their completion, distributed across the months of the year.

A	B	C	D	E	F	G	H	I	J	K	L	AI	AJ	AK	AL	AM
SUPERINTE	DEPARTAM	DIVISÃO	CATEGORIA	ÁREA	PROJETO	SUB PROJETO	CONTRATO	PEP	COLABORA	REGISTRO	Atividade	jan/23	fev/23	mar/23	abr/23	mai/23
SEP	DGET	VESE	PROJETO GET	COPEL GET	RPPM25	JGI-MEL12			I-21-15165301	FABIOLA AZUN	51727 ANÁLISE DE PROJETO CIVIL OU GEOLOGIA OU	6	6	6	6	6
SEP	DGET	VESE	PROJETO GET	COPEL GET	RPPM25	UBR-MEL18			I-21-15163101	FABIOLA AZUN	51727 ANÁLISE DE PROJETO CIVIL OU GEOLOGIA OU	6	6	6	6	6
SEP	DGET	VESE	INSTITUCIONA	COPEL GET	TREINAMENTC	TREINAMENTOS ANUAIS RELACIONADOS A ATIVID				FABIOLA AZUN	51727 PARTICIPAÇÃO DE TREINAMENTO	8	8	8	8	8
SEP	DGET	VESE	INSTITUCIONA	COPEL GET	TREINAMENTC	SS				FABIOLA AZUN	51727 PARTICIPAÇÃO DE ROTINA DA EMPRESA	2	2	2	2	2
SEP	DGET	VESE	INSTITUCIONA	COPEL GET	TREINAMENTC	PADRONIZAÇÃO PROJETO PADRÃO				FABIOLA AZUN	51727 ELABORAÇÃO DE TREINAMENTO	15	15	15	15	15
SEP	DGET	VESE	PROJETO GET	COPEL GET	ACESSANTE	GUAÍRA TRANSMISSORA				FELIPE LEMOS	49738 ANÁLISE DE EQUIPAMENTOS E MATERIAIS	0	0	0	0	0
SEP	DGET	VESE	PROJETO GET	COPEL GET	ACESSANTE	ENGIE				FELIPE LEMOS	49738 ANÁLISE DE EQUIPAMENTOS E MATERIAIS	0	0	0	0	0
SEP	DGET	VESE	PROJETO GET	COPEL GET	ACESSANTE	ELETROSUL				FELIPE LEMOS	49738 ANÁLISE DE EQUIPAMENTOS E MATERIAIS	1	1	1	1	1
SEP	DGET	VESE	PROJETO GET	COPEL GET	ACESSANTE	NEOENERGIA-CCI	COPEL GET Nº 00 S-20-0020600			FELIPE LEMOS	49738 ANÁLISE DE EQUIPAMENTOS E MATERIAIS	1	1	1	1	1
SEP	DGET	VESE	PROJETO GET	COPEL GET	ACESSANTE	AIE		S-19-9013670		FELIPE LEMOS	49738 ANÁLISE DE EQUIPAMENTOS E MATERIAIS	1	1	1	1	1
SEP	DGET	VESE	PROJETO GET	COPEL GET	ACESSANTE	PCH TITO				FELIPE LEMOS	49738 ANÁLISE DE PROJETO ELÉTRICO	4	4	4	4	4
SEP	DGET	VESE	INSTITUCIONA	COPEL GET	FÉRIAS					FELIPE LEMOS	49738 PARTICIPAÇÃO DE ROTINA DA EMPRESA	14	14	14	14	14
SEP	DGET	VESE	INSTITUCIONA	COPEL GET	PESSOAL	ROTINA DA EMPRESA				FELIPE LEMOS	49738 PARTICIPAÇÃO DE ROTINA DA EMPRESA	4	4	4	4	4
SEP	DGET	VESE	PROJETO GET	COPEL GET	REA 10688	CIC-AMP21		I-21-15285301		FELIPE LEMOS	49738 GESTÃO/COORDENAÇÃO DE DOCUMENTOS T	0	0	0	0	0
SEP	DGET	VESE	PROJETO GET	COPEL GET	REA 10688	CIC-AMP21		I-21-15285301		FELIPE LEMOS	49738 ANÁLISE DE PROJETO ELÉTRICO	12	12	12	12	12
SEP	DGET	VESE	PROJETO GET	COPEL GET	REA 7384	IBP-REF01	4600020052	I-16-67576301		FELIPE LEMOS	49738 ANÁLISE DE PROJETO ELÉTRICO	0	0	0	0	0
SEP	DGET	VESE	PROJETO GET	COPEL GET	REA 7384	LON-REF01	4600020052	I-18-85641601		FELIPE LEMOS	49738 ANÁLISE DE PROJETO ELÉTRICO	0	0	0	0	0
SEP	DGET	VESE	PROJETO GET	COPEL GET	REA 9219	GUA-AMP21	4600023116	I-20-05365801		FELIPE LEMOS	49738 ELABORAÇÃO DE DOCUMENTOS TÉCNICOS (IN	0	0	0	0	0
SEP	DGET	VESE	PROJETO GET	COPEL GET	REA 9219	GUA-AMP21	4600023116	I-20-05365801		FELIPE LEMOS	49738 ANÁLISE DE PROJETO ELÉTRICO	12	12	12	12	12
SEP	DGET	VESE	PROJETO GET	COPEL GET	REA 9219	GUA-MEL06	4600023116	I-21-15017201		FELIPE LEMOS	49738 ELABORAÇÃO DE DOCUMENTOS TÉCNICOS (IN	0	0	0	0	0
SEP	DGET	VESE	PROJETO GET	COPEL GET	REA 9219	GUA-MEL06	4600023116	I-21-15017201		FELIPE LEMOS	49738 ANÁLISE DE PROJETO ELÉTRICO	12	12	12	12	12
SEP	DGET	VESE	PROJETO GET	COPEL GET	REA 9219	GUA-AMP21	4600023116	I-20-05365801		FELIPE LEMOS	49738 ANÁLISE DE DOCUMENTOS TÉCNICOS (INCLUS	0	0	0	0	0
SEP	DGET	VESE	PROJETO GET	COPEL GET	REA 9219	GUA-AMP21	4600023116	I-20-05365801		FELIPE LEMOS	49738 ANÁLISE DE EQUIPAMENTOS E MATERIAIS	0	0	0	0	0
SEP	DGET	VESE	PROJETO GET	COPEL GET	REA 9564	SDI-AMP12	4600024366	I-20-05481201		FELIPE LEMOS	49738 ANÁLISE DE DOCUMENTOS TÉCNICOS (INCLUS	0	0	0	0	0
SEP	DGET	VESE	PROJETO GET	COPEL GET	REA 9564	SDI-AMP12	4600024366	I-20-05481201		FELIPE LEMOS	49738 ANÁLISE DE EQUIPAMENTOS E MATERIAIS	0	0	0	0	0
SEP	DGET	VESE	PROJETO GET	COPEL GET	RPPM13	CCO-MEL07	4600018530	I-18-85249201		FELIPE LEMOS	49738 ANÁLISE DE PROJETO ELÉTRICO	0	0	0	0	0

Figure 3. Planned hours recording spreadsheet.

It is important to highlight that, as there was no historical database for recording planned hours for activity execution, the effort estimates recorded in the planning spreadsheets were filled out by the area managers and supervisors based on expert opinions and analogous estimates.

Organization of information

During the information and data source mapping stage, it was found that organizing the information would be a challenge, as dozens of parallel controls were identified, including spreadsheets, forms, schedules, and email routines. To address this issue, it was decided to organize the input of information through the creation and use of Microsoft Lists. The lists were developed on the SharePoint site of Copel GeT's project office, organized by topic, and structured using a relational model. An example is presented in the Figure 4.

SharePoint

Pesquisar nesta lista

+ Novo Editar no modo de exibição de grade Compartilhar Exportar Forms Automatizar Integrar ...

Todos os Itens

BD_PMO_GET - Lista de Contratos ☆

ID	Nº do Contrato	Fornecedor	Nome Atual da ...	Objeto	Status do Contr...	Projeto	Gestor	Exibir no BI	Objeto Detalhado	Data de Assinatura
1	4600016882	KRUK CONSULTORIA, AVALIACOES E PERI		Fundiário PCH Bela Vista	Concluído	PCH Bela Vista	Roberto Werneck Seara	Sim	Prestação de serviços de engenharia - para demarcação, avaliação, liberação e regularização fundiária, bem como serviços correlatos, da PCH Bela Vista e Sistema de Transmissão Associado, empreendimentos componentes do Leilão A-6	27/02/2019
2	4600017235	SOMA SERVICOS ORGANIZACAO E MEIO		Meio Ambiente PCH Bela Vista	Em Andamento	PCH Bela Vista	Roberto Werneck Seara	Sim	Licenciamento ambiental para PCH Bela Vista, incluindo o acompanhamento da emissão da licença de instalação, o monitoramento e/ou execução dos programas ambientais previstos no Projeto Básico Ambiental e a obtenção da licença de operação da PCH Bela Vista, além dos serviços de licenciamento	22/04/2019
3	4600017311	DELOITTE TOUCHE TOHMATSU		Auditoria Externa	Concluído	PCH Bela Vista		Não	contrato cancelado	02/05/2019
4	4600017497	WEG Equipamentos		5x Transf. 230 kV - 150 e 50 MVA	Concluído	Reforços SE1	Rafael Alves Pereira	Sim	Fornecimento, montagem, comissionamento de transformador trifásico SE Cascavel 230kV	13/06/2019
5	4600017582	CONCRETIZAR ENGENHARIA DE OBRAS LTD		EPC PCH Bela Vista	Em Andamento	PCH Bela Vista	Roberto Werneck Seara	Sim	Implantação da PCH BELA VISTA no Rio Chopim, no município de Verê, por ambas as margens, no Estado do Paraná, mediante empreitada sob o regime de empreitada por preço global, com o fornecimento global dos bens e prestação dos serviços necessários para colocar o empreendimento em plena	29/05/2019
6	4600017764	WEG Equipamentos		4x Transf. 230 KV - 225 MVA	Concluído	Reforços SE1	Rafael Alves Pereira	Sim	Autotransformador Trifásico - SE Ponta Grossa Sui 230kV ATF1 // Autotransformador Trifásico - SEPonta Grossa Sui 230 kV ATF2 // Autotransformador Trifásico - SE Pato Branco 230 kV ATF1 // Autotransformador Trifásico - SE Pato Branco 230 kV ATF2	06/08/2019
7	4600017780	COPEL GERACAO E TRANSMISSAO S.A		EP Bela Vista	Em Andamento	PCH Bela Vista	Roberto Werneck Seara	Sim	Engenharia do proprietário para acompanhamento técnico e fiscalização da execução do Contrato EPC, no que diz respeito ao Projeto, implantação das obras civis, fabricação, montagem e comissionamento dos equipamentos eletromecânicos de Bela Vista	26/07/2019
8	4600018110	CTV		Melhoria SE Cascavel 500kV	Em Andamento	SE1	Guilherme Marques Dos S	Sim	Melhoria SE Cascavel 500kV, Bela Branca 320 kV, Fátima	12/09/2019

Contagem
116

Figure 4. Planned hours recording spreadsheet.

The data was grouped by type and use, where each type has a list for organizing the information. The grouping strategy adopted is shown in Table 2.

Table 2. Mapped essential information and respective data sources.

Grouping Information	Characteristics of the list
Basic Data	such as scope, classification, project type, start and end dates, project manager, baseline CAPEX value, contingency percentage, estimated revenue value, ANEEL deadline, internal rate of return of the project, etc.
Physical progress	organizes the control records of the physical progress of projects, planned, realized and forecast percentages.
Financial Progress	control records of the financial progress of projects, planned and realized monetary values and future disbursement forecasts.
Risks	has records of the risks identified for each project, their respective status attributes, probability and impact estimates, responsible parties, action plans and occurrence records.
Contracts	has a list of all the contracts that make up each of the projects. This list serves as a reference for the automation of some of the SAP reports, in which data relating to the contract is extracted, such as the contract balance, payments made to the supplier, contract amendments, etc.
Cost Collectors	has a list of all the cost accounting collectors for each of the projects and budget programs. This list also serves as a reference for SAP report automations, in which data is extracted from the accounting entries for each project, such as costs for own labor, travel, etc.

Relational Data Model

With the data structure defined, the development of the Extraction, Transformation, and Load (ETL) stage began using the Power Query solution in Microsoft Power BI. The ETL is a fundamental process in the field of data management, used to integrate and consolidate information from different sources into a single destination.

After implementing the ETL stage, the dataset in Power BI was developed by constructing the relational model between the various tables derived from the data flow outputs. Figure 5 illustrates the relational model developed in the Power BI dataset.

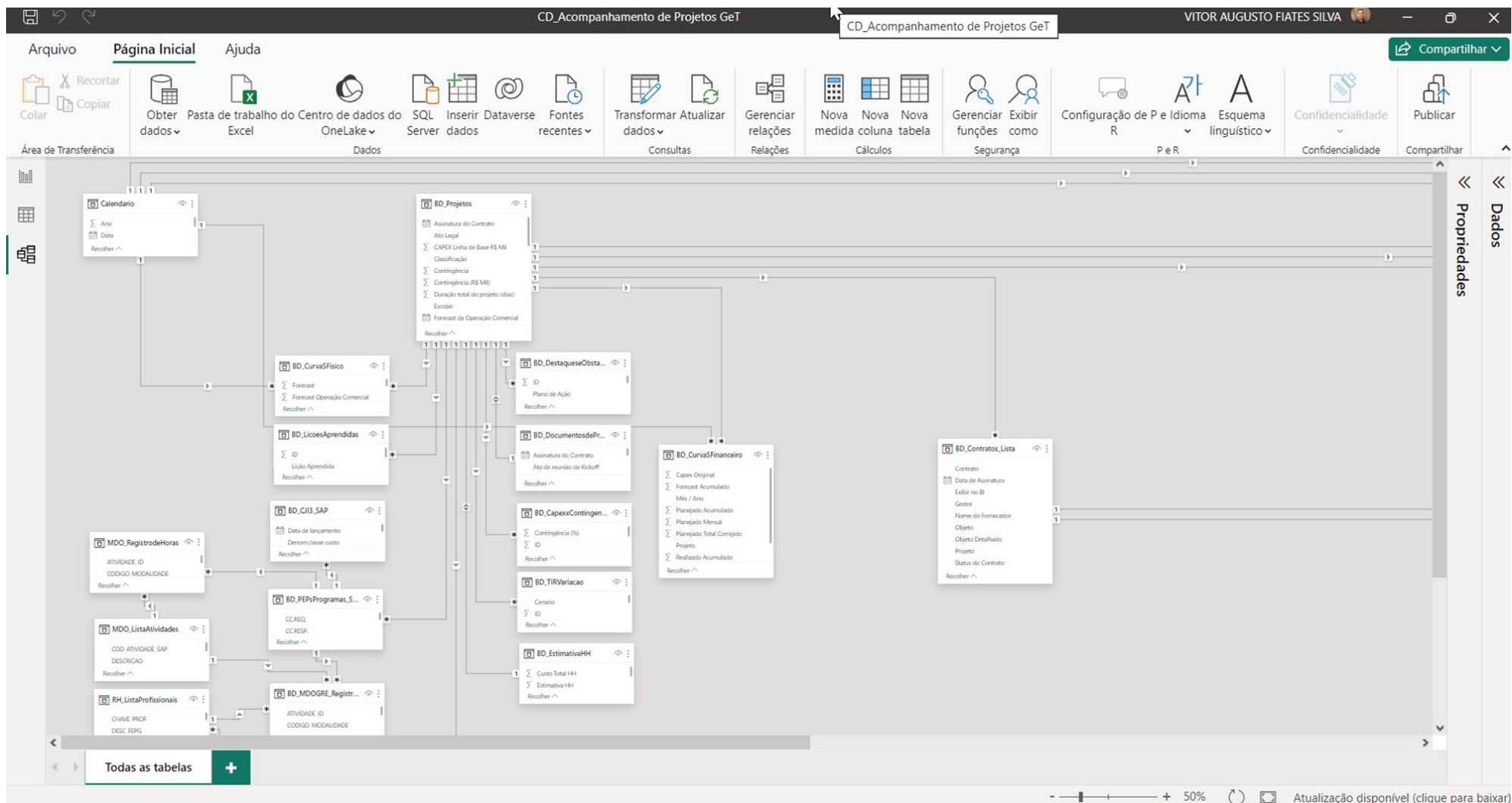


Figure 5. Sample of the relational model built on the dataset.

After the development of the dataset, Power BI was used to create interactive reports, also known as dashboards, for tracking, monitoring, and controlling transmission projects and works. The selection of graphical components, layout definition, and adjustments to the reports were carried out based on requirements gathered during meetings with the technical teams from Copel GeT. Additionally, the performance indicators to be used for assessing resource allocation efficiency and project performance were defined. The resultant dashboards are presented in next section.

RESULTS AND DISCUSSIONS

The purpose of this chapter is to analyze how the data and business intelligence methodology applied to transmission projects and construction organizes the information, allowing for the evaluation of the physical and financial performance of the projects, as well as resource allocation. This contributes to improving management, decision-making, and, consequently, the results achieved by the organization.

The organization of data in a collaborative and structured database enabled the development of a set of dashboards that integrate several pieces of information from the transmission projects and construction at Copel GeT. In total, two dashboards were developed, each with specific scope and objectives, based on company management experience as listed in the following items:

- *Project Monitoring Dashboard*: Its main objective is to assist managers and their teams in the implementation management by monitoring physical progress, financial status, risk management, and tracking the hours worked by team members.
- *Planning and Man-Hours Reconciliation Dashboard*: Its goal is to help managers and coordinators manage technical teams by monitoring the execution of the planned allocation of activities, as well as the performance of team execution.

Transmission reinforcement and improvement projects at Copel GeT are organized into projects, which consist of one or more construction works, according to criteria such as location, execution timeline, legal authorization, complexity, and synergy of shutdowns.

The developed methodology enabled the organization of project data in a structured manner, centralizing and standardizing the information, so that the monitoring and implementation management reports were presented in a standardized way through the dashboards, which are detailed below.

Basic Data Report

The report titled "Basic Data" provides a summary of the main project information, such as project name, project manager, start date, end date, project status, location, project margin (internal rate of return), Capex, scope, status of project documentation preparation, key highlights, and obstacles identified throughout the project. Figure 6 illustrates an example of the "Basic Data" report. The example refers to a set of reinforcement works carried out at the CIC substation in the city of Curitiba.

Reforços SE4 - CIC

Dados Básicos



Gestor

Rafael Alves Pereira

Início

2021-09

Término

2024-11

Status

Em Execução

Local

Curitiba - PR

Margem (%)

12,38

Escopo Básico

Instalação de transformador trifásico 230/13,8 kV; Instalação de módulos de conexão de 230 kV e 13,8 kV e Adequação...

CAPEX (R\$ mil)

Linha Base

31.212

Nominal

33.560

Forecast

33.354

Documentos

TAP

Designação RH

Reunião Kickoff

PGP

TM

TEP

Destaques

Antecipação das receitas
Reconhecimento de receita adicional

Obstáculos

Ainda não foi publicado edital da DIS – previsão continua sendo maio/23 para o contrato deles.
Concorrência de mão de obra interna Copel
Incompatibilidades na capacidade da proteção de barras 230 da SE foram identificados e a engenharia está buscando uma alternativa técnicas

Figure 6. Basic Data Report.

The "Basic Data" report helps managers and their teams in presenting the project during periodic status report meetings with the company's executive board, and contributes to the easy access to information about the projects that make up the company's portfolio.

S-Curve Reports

The S-curve reports are divided into two: the physical S-curve and the financial S-curve. Both are presented in Figure 7.

The report titled "Physical S Curve" presents the planned and actual physical progress of the project, as shown in Figure 7(a). The curves represent the cumulative series of the planned and actual progress percentages month by month, while the vertical bars represent the monthly increments. This graph also displays key milestones, including the planned commercial operation start date and the deadline set by ANEEL for project completion.

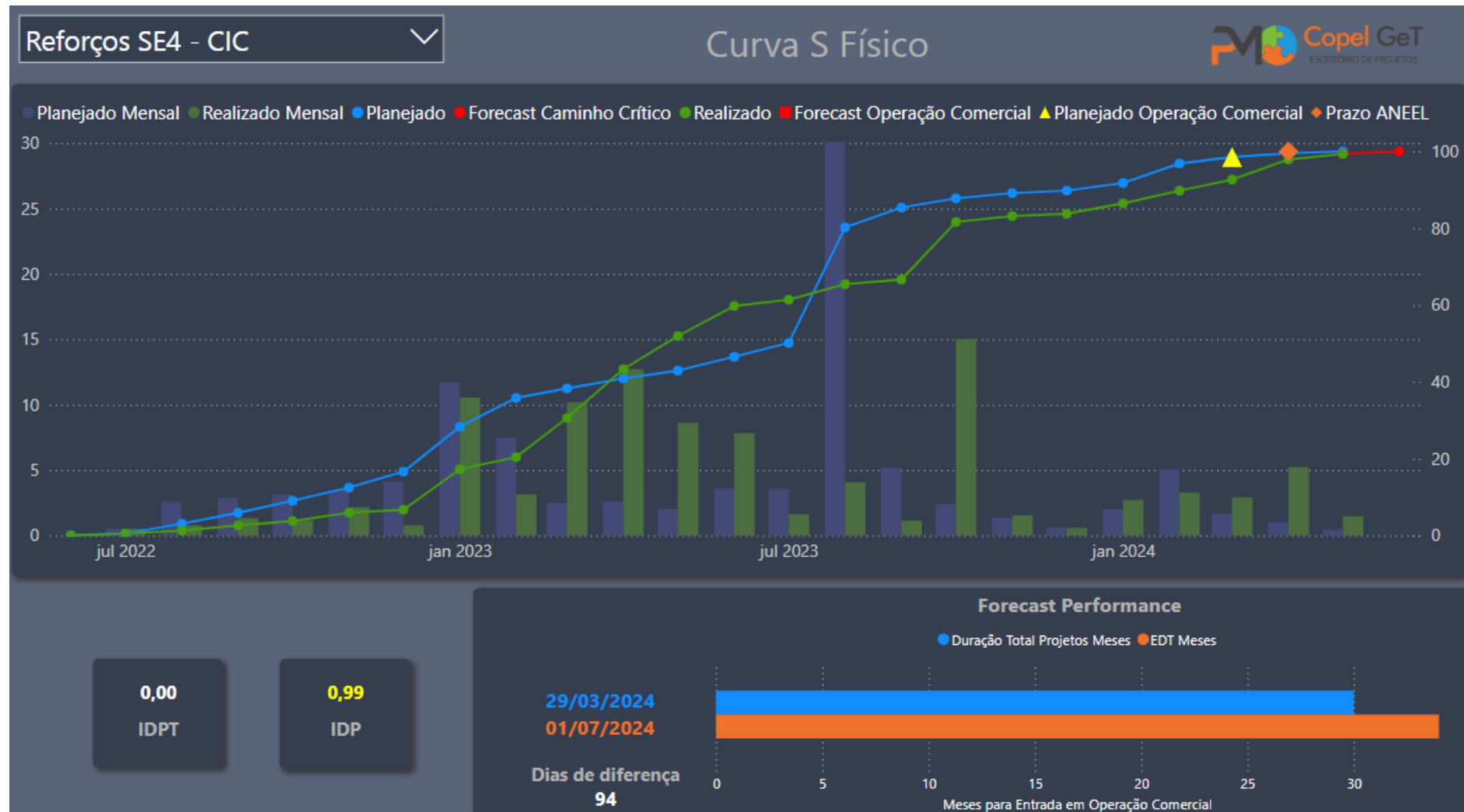
The cards located at the bottom left corner of the report present the Earned Value Index (EVI), the Performance Index for Completion (IDPT), and the Schedule Performance Index (IDP), which help managers monitor the project's execution performance. The horizontal bar chart located at the bottom right corner shows the projected project completion, calculated based on the Schedule Performance Index (IDP). The "Physical S Curve" report helps managers and their teams monitor and present the physical progress of their projects, particularly during the periodic status report meetings with the company's executive board.

The report titled "Financial S Curve" presents the planned and actual financial progress of the project, as shown in Figure 7(b). The curves represent the cumulative series of the planned and actual financial values month by month, while the vertical bars represent the monthly increments. This graph also displays the projected Capex for project completion, calculated by applying a financial adjustment to the remaining balance to be disbursed, using the IPCA inflation index.

The card located in the upper left corner of the report presents the Cost Performance Index (CPI), calculated by the earned value, helping the manager monitor the financial execution performance of the project. Therefore, the "Financial S Curve" report supports managers and their teams in monitoring the cash flow curve and presenting the financial progress of their projects during periodic status report meetings with the company's executive board.

The report titled "Physical S Curve" presents the planned and actual physical progress of the project, as shown in Figure 7(a). The curves represent the cumulative series of the planned and actual progress percentages month by month, while the vertical bars represent the monthly increments. This graph also displays key milestones, including the planned commercial operation start date and the deadline set by ANEEL for project completion.

(a)



(b)

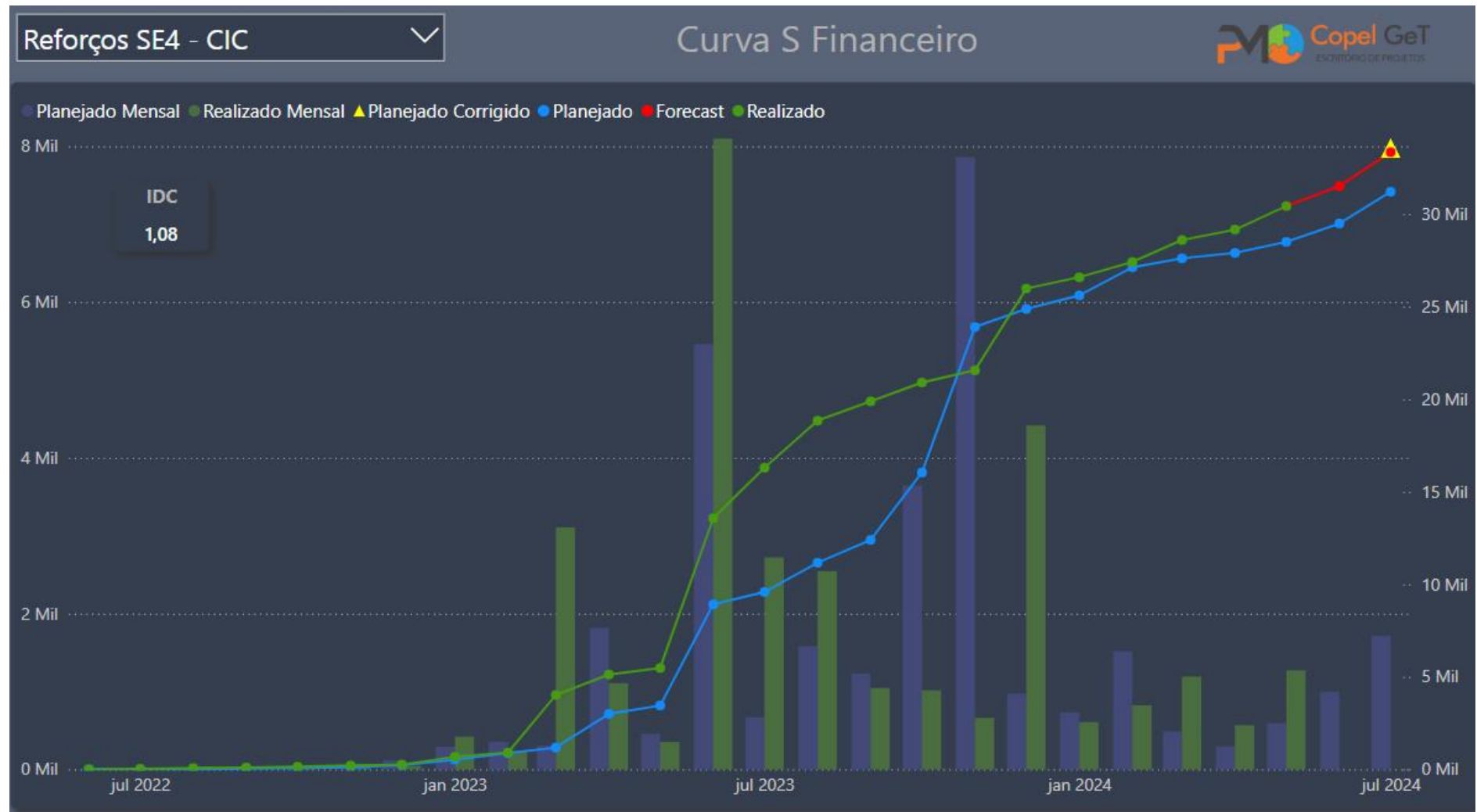


Figure 7. (a) Physical and (b) Financial S-Curve Reports.

The cards located at the bottom left corner of the report present the Earned Value Index (EVI), the Performance Index for Completion (IDPT), and the Schedule Performance Index (IDP), which help managers monitor the project's execution performance. The horizontal bar chart located at the bottom right corner shows the projected project completion, calculated based on the Schedule Performance Index (IDP). The "Physical S Curve" report helps managers and

their teams monitor and present the physical progress of their projects, particularly during the periodic status report meetings with the company's executive board.

The report titled "Financial S Curve" presents the planned and actual financial progress of the project, as shown in Figure 7(b). The curves represent the cumulative series of the planned and actual financial values month by month, while the vertical bars represent the monthly increments. This graph also displays the projected Capex for project completion, calculated by applying a financial adjustment to the remaining balance to be disbursed, using the IPCA inflation index.

The card located in the upper left corner of the report presents the Cost Performance Index (CPI), calculated by the earned value, helping the manager monitor the financial execution performance of the project. Therefore, the "Financial S Curve" report supports managers and their teams in monitoring the cash flow curve and presenting the financial progress of their projects during periodic status report meetings with the company's executive board.

CAPEX Reports

The CAPEX (capital expenditure) reports are organized into two: the actual capex report presented in Figure 8(a) and the capex, margin and contingency report presented in Figure 8(b).

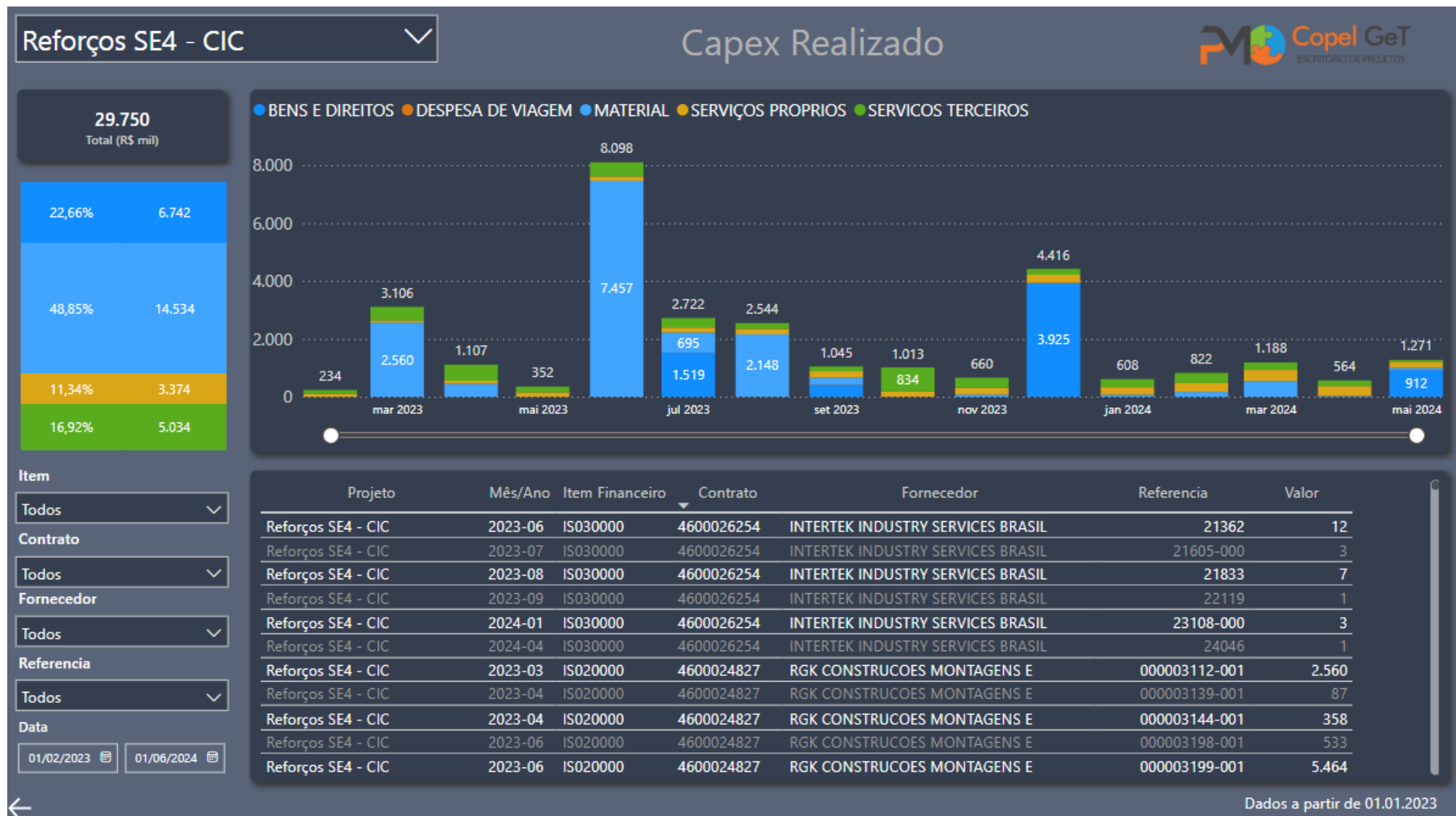
The report titled "Realized Capex " presents detailed information on the monthly disbursements made in the project, as shown in Figure 7(a). The vertical bar chart located in the top-left corner displays the accumulated financial values in the categories of assets and rights, travel expenses, materials, own services, and third-party services. This chart also shows the percentage share of each category relative to the total disbursed amount.

The vertical bar chart in the center of Figure 7(a) presents the breakdown of monthly disbursement values. These records are detailed in the table located at the bottom of the report. Additionally, the segmentation fields in the bottom-left corner of the report help managers in the selection and search of specific disbursement records. This report provides managers with detailed access to the disbursements of their projects and a breakdown of cost compositions. This characterization enables the establishment of comparative benchmarks between similar projects within the company.

The report titled "Capex x Contingency x Margin" presents a sensitivity analysis of the variations affecting the project's margin, considering the impacts of the macroeconomic scenario, the actual disbursement curve, and the early or delayed operational startup, as shown in Figure 7(b). The graph at the bottom left of the report displays the percentage impact of the macroeconomic scenario, disbursement curve, and operational startup on the project's initial margin. The graph at the top of the report shows the financial progress curve of the project, through which temporal and amplitude variations in the disbursement curve can be identified.

The cards located in the bottom right corner of the report display the project's forecasted and actual contingency values, allowing identification of how much of the predicted value has already been used. The unused contingency contributes to the project's margin increase. The "Capex x Contingency x Margin" report helps managers present the financial results of the project during the periodic status report meetings with the company's board of directors.

(a)



(b)

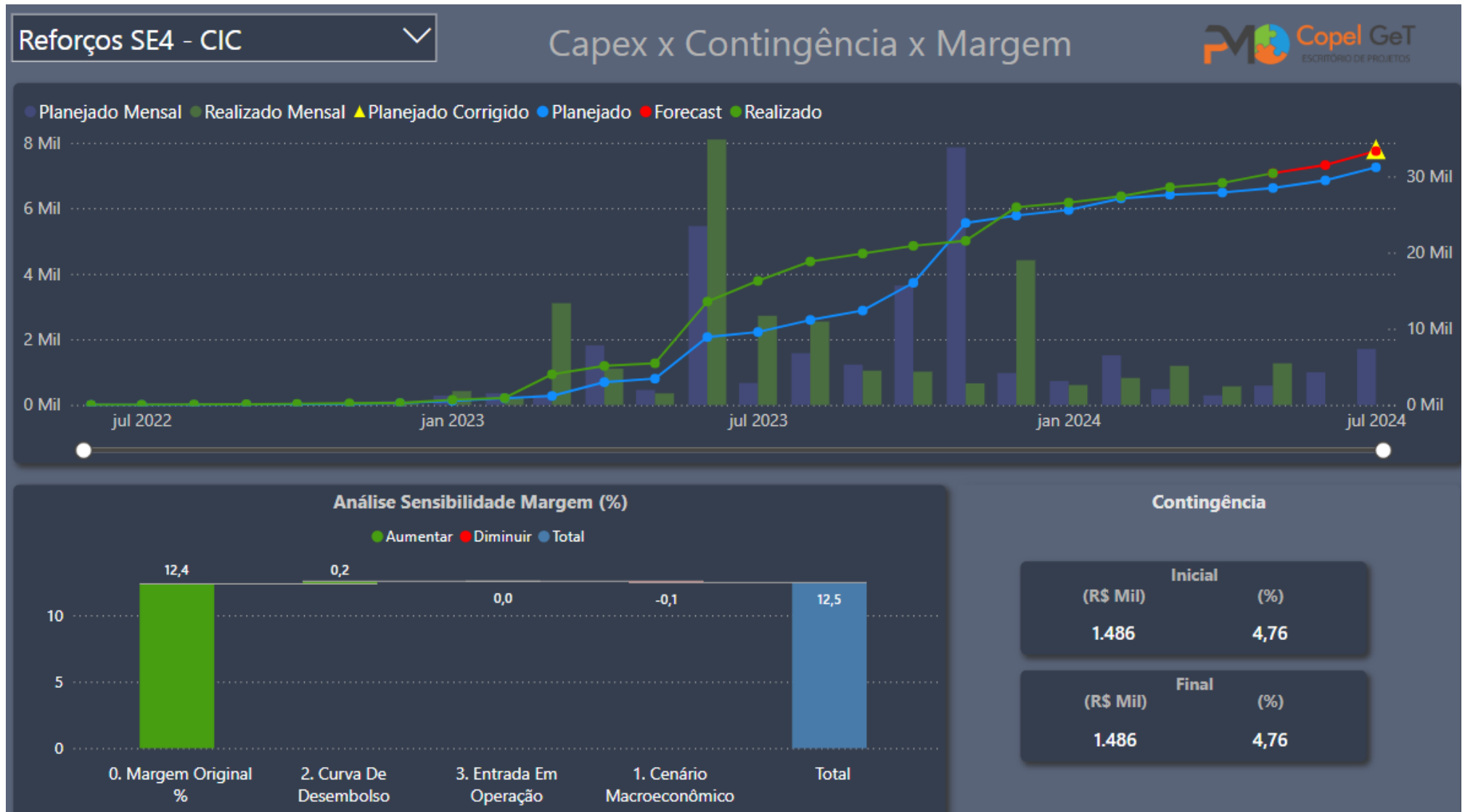


Figure 8. (a) Actual CAPEX report and (b) CAPEX, margin and contingency report.

Risk Report

The report titled "Riscos" presents the identified risks, both qualitative and quantitative assessments, the available contingency balance, and records of risk materialization events throughout the project lifecycle, as shown in Figure 9.



Figure 9. Risk Reports.

Note: Probability vs. Impact; Class x VME (Expected Monetary Value); Risk Breakdown.

The bubble chart located in the central part of the report displays the qualitative risk assessment, using five-level scales for probability and impact. The size of the bubbles is determined by the severity of the risk, which is the product of the probability and impact scale values. The colors of the bubbles represent the classification used for risk categorization.

The tornado chart on the right side of the report displays the ranking of risk categories based on their Expected Monetary Value (VME). The table at the bottom of the report lists the identified risks along with their assessments of probability, impact, Expected Monetary Value (VME), counters for the number of occurrences of materializations, and recorded impacts. The VME is the financial value resulting from the product of probability and impact.

In the left side, slicer boxes allow managers and their teams to filter by the projects involved, risk categories, risk status (under monitoring, materialized, mitigated), and impact categories (schedule, cost, or both schedule and cost). The cards in the top right corner of the report show the total estimated impact of the risks, the available contingency balance, the sum of the Expected Monetary Values, and the total number of identified risks.

The "Risks" report supports managers and their teams in the risk management process throughout the project lifecycle. Through this report, project managers can monitor risks and highlight the most relevant events to address mitigation actions. This report also allows for reviewing the history of risk materializations, serving as a database for future studies of similar works and projects.

Man-Hour Management Report

The report titled "Man-Hour Management" presents the allocation of human resources throughout the project lifecycle, as shown in Figure 10. The vertical bar chart located at the top of the report displays the monthly time entries recorded by team members who worked on project activities.

On the right side of the screen, the table in the upper corner displays the list of professionals who worked on the project activities, along with their respective departments, total hours worked, and the percentage of their hours in relation to the total recorded. This report allows for an assessment of the effort contributed by each allocated professional. Still on the right side, the donut chart provides a summary of the hours logged by each company department. This report helps identify which areas were most involved in the project's execution.

Finally, the horizontal bar chart in the lower corner summarizes the hours logged by cost collector. This chart enables managers to monitor the distribution of time entries across the various works that may comprise the project, as multiple cost collectors are often involved in such cases.

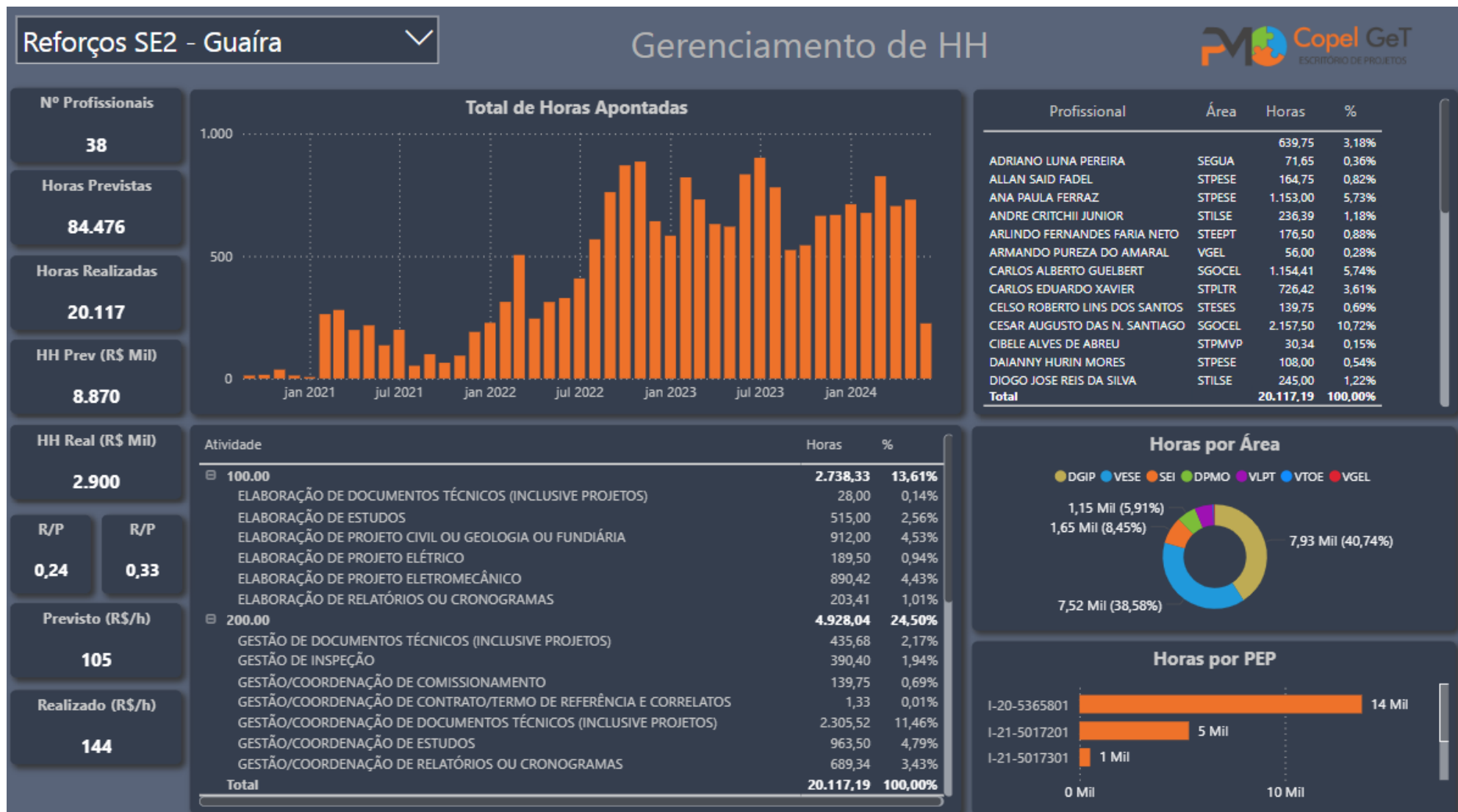


Figure 10. Man-Hour Management Reports.
Note: total hours reported; employees; activity; hours engaged per unit.

Results Analysis

The data and business intelligence methodology in the project management process was used to support the management of large-scale reinforcement projects T2021-074, T2019-084, T2020-074, T2019-086, and T2019-085, which were authorized by ANEEL through the authorization resolutions REA 10688/2021, REA 9741/2021, REA 9564/2020, and REA 8543/2020. The Table 3 compiles the performance results for the execution of each project based on of schedule and budget. It shows real gains in project completion times and the resulting financial return.

Table 3. Practical cases and observed impacts.

Construction	On-Time Performance	On-Budget Performance
T2021-074 - SE Cidade Industrial de Curitiba - Installation of the 3rd Transformer 230/13,8 kV - 50 MVA.	6 days ahead of the regulatory deadline.	Savings of R\$ 2.5 million in relation to planned Capex.
T2019-084 - Recapacitation of LT 230 kV Campo Comprido - Santa Quitéria, C1.	97 days ahead of the regulatory deadline.	Savings of R\$ 1.49 million in relation to planned Capex.
T2020-074 - Installation of the 3rd Three-Phase Autotransformer 230/138 kV – 150 MVA SE Sarandi	59 days ahead of the regulatory deadline.	Savings of R\$ 50.000 compared to the planned Capex.
T2019-086 - Recapacitation of LT 230 kV Pilarzinho - Santa Monica C1	27 days behind the regulatory deadline.	Savings of R\$ 5 million in relation to planned Capex.
T2019-085 - Recapacitation of LT 230 kV Bateias -Pilarzinho	153 days ahead of the regulatory deadline.	Savings of R\$ 5 million in relation to planned Capex.

The analysis of Table 3 shows that the implementation of the data and business intelligence methodology in the project management process contributed to improved execution performance in these cases of reinforcement projects. This outcome is further supported by data published by the National Electric Energy Agency (ANEEL) in the transmission project information panel [11] as shown in the Figure 11.

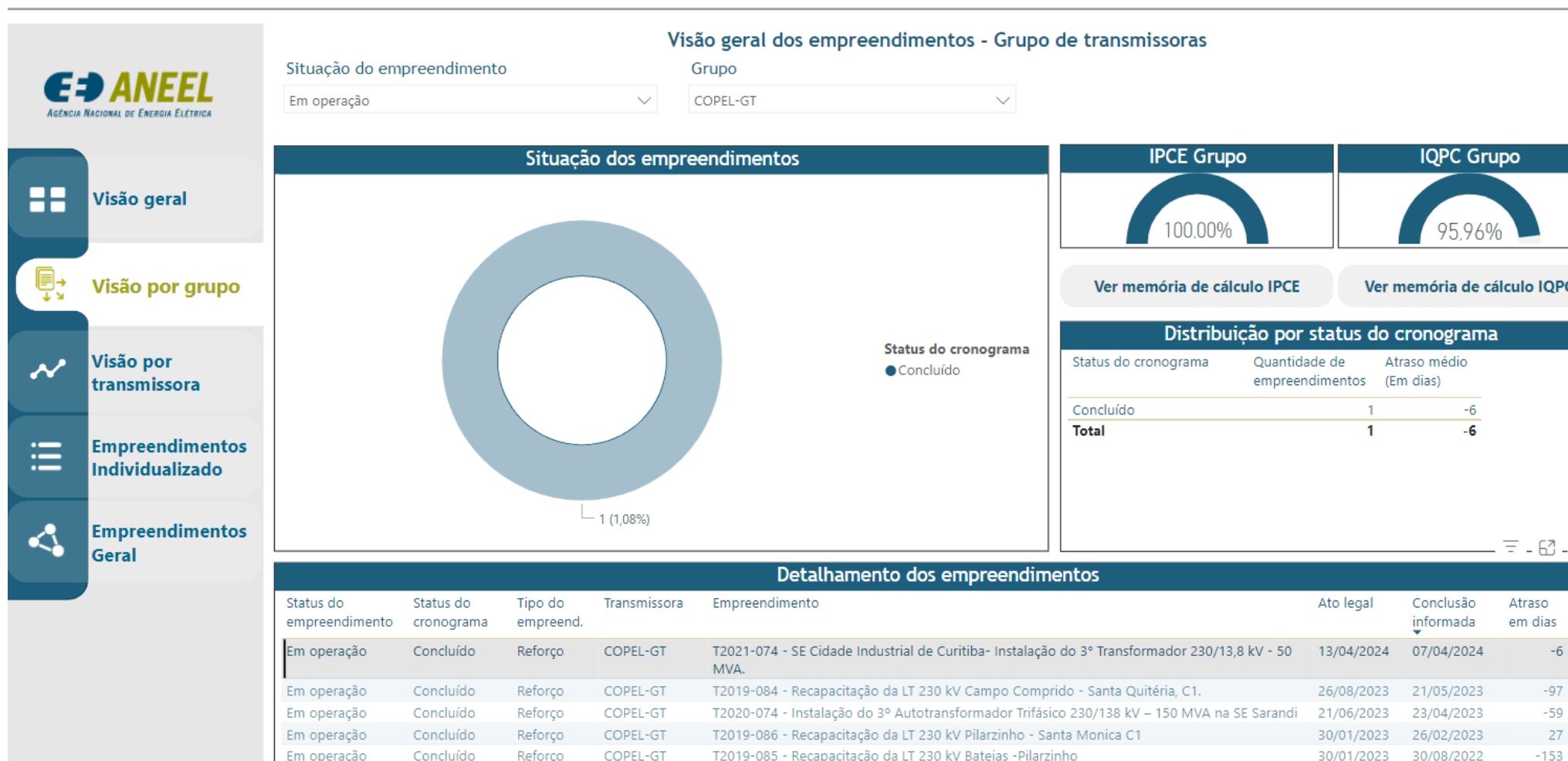


Figure 11. Information panel on transmission projects.

One of the key advancements observed with the implementation of the methodology was the early delivery of projects and stakeholder satisfaction with the standardization and centralization of information. Additionally, the methodology contributed to risk management, which enabled a reduction in capital expenditure (Capex), resulting in significant savings in project budgets. This financial efficiency was accompanied by an optimization in the allocation of human resources, with a better utilization of available skills. The strategic alignment between data intelligence and business management provided a more integrated and dynamic view of operations, enhancing results and setting a new standard of efficiency in project management.

CONCLUSION

This work aimed to develop a business intelligence and data methodology with respective software tools, to support the efficient management process of projects in the electric power sector, particularly in energy transmission projects. The set of management reports developed on a business intelligence platform proved to be an effective tool for project management, providing analyses on the efficiency of resource allocation within the company.

The methodology was built based on a data organization, extraction, transformation, and loading (ETL) framework, enabling integration and availability of information from various company systems. Furthermore, the structuring of a relational data model allowed the development of management reports that provide qualified information, essential for project management.

The creation of performance indicators for monitoring activities and the qualification of extracted information led to the development of a data foundation for strategic decision-making by managers. The management reports developed in the business intelligence tool were able to provide visibility into project progress, resource allocation, and operational efficiency, thus meeting the proposed objectives.

Therefore, this work directly contributes to the improvement of the project management process in the electric power sector, introducing innovation through the application of business intelligence and data analytics. The results obtained demonstrate that the developed methodology has the potential to become a strategic tool for companies in the sector, providing efficiency gains and greater control over the resources employed in transmission projects.

Certainly, there are limitations to the full adoption and effectiveness of the proposed methodology. Three main aspects show these limitations: data availability, real adoption and implementation of the method and the degree of maturity of the data culture within the company. The first item, although obvious and critical, is sometimes neglected. The basis of the methodology is positioned over good information structure and data organization. The second aspect signals the organization's real desire to value its project management structure. Finally, a good level of data culture can greatly help in implementing new methodologies in the company. Without these three aspects well established, the proposed method cannot be fully implemented, nor can it present satisfactory results.

As a suggestion for future work, it is recommended to expand this methodology to other types of projects in the electric power sector, such as power generation projects. Additionally, the adoption of more advanced technologies, such as machine learning and artificial intelligence, could further enhance the analytical capabilities of the management reports.

Funding: This work was financed by the program for research and technological development of Companhia Paranaense de Energia - COPEL, through the project PD-06491-0565/2019, regulated by Agência Nacional de Energia Elétrica - ANEEL.

Acknowledgments: The authors would like to thank COPEL and ANEEL for the opportunity and support.

Conflicts of Interest: The authors declare no conflict of interest.

Data Availability Statement: Research data are not available.

REFERENCES

1. ANEEL. [Normative Resolution nº 905] Resolução Normativa nº 905. Diário Oficial da União. 2020; 158(241)
2. Lopes AB, Boscardioli C. Business intelligence and analytics to support management in construction: a systematic literature review. Rev Bra Comp Apl. 2021;13(1):27-41.
3. Gupta U, Cannon S. A Practitioner's Guide to Data Governance. Emerald Publishing Limited; 2020.
4. Chae BK, Olson DL. 2013. Int. J. Inf. Technol. Decis. Mak. 2013; 12: 9-26.
5. Wixom B, BruceY, Relich M 2013. Maximizing Value from Business Analytics. MIS Quarterly Executive. 2013; 12(2):111-23.
6. Sharda R, Asamoah D, Ponna N. 2013. Research and Pedagogy in Business Analytics: Opportunities and Illustrative Examples. J Comp Inf Tech. 2013; 21(3): 171-83.
7. Chater N, Tenenbaum JB, Yuille A. 2006. Probabilistic models of cognition: Conceptual foundations. Trends Cogn Sci. 2006; 10(7): 287-91.
8. Loshin D. Enterprise Knowledge Management: The Data Quality Approach (The Morgan Kaufmann Series in Data Management Systems). Morgan Kaufmann; 2001.
9. Marshall L, Harpe D. 2009. Decision making in the context of business intelligence and data quality. South African J Inf Manag; 11(2);

10. Debbarma N, Nath G, Das H. 2013. Analysis of Data Quality and Performance Issues in Data Warehousing and Business Intelligence. *Int J Comp Applic*; 79(15): 20-6.
11. ANEEL. Agência Nacional de Energia Elétrica, [Transmission Project Information Panel] "Painel de Informações de Empreendimentos de Transmissão", <https://portalrelatorios.aneel.gov.br/trans/trans>



© 2025 by the authors. Submitted for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>)